



Graph Neural Networks

Graph Convolutional Networks

Teaser

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Graph Neural Networks

$$\mathbf{H}^{t+1} = F(\mathbf{H}^t, \mathbf{X})$$

F propagation step



"Simple" GCN

$$\mathbf{H}^{t+1} = \sigma(\mathbf{W}^t \mathbf{A} \mathbf{H}^t)$$

$\mathbf{A}$ adjacency matrix

σ e.g. ReLU



"Simple" GCN

"messages"
from neighbours



$$H^{t+1} = \sigma(W^t A H^t)$$

A adjacency matrix

σ e.g. ReLU



Self Loops

self loops

$$\mathbf{H}^{t+1} = \sigma(\mathbf{W}^t \hat{\mathbf{A}} \mathbf{H}^t)$$

$$\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}$$

$\mathbf{I}$ identity matrix



Graph Convolutional Networks

self loops

$$\mathbf{H}^{t+1} = \sigma(\mathbf{W}^t \underbrace{\hat{\mathbf{D}}^{-\frac{1}{2}} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-\frac{1}{2}}}_{\text{normalization}})$$

normalization

$\hat{\mathbf{D}}$ Diagonal node degree matrix of $\hat{\mathbf{A}}$



"Simple" GCN

$$\mathbf{H}^{t+1} = \sigma(\mathbf{W}^t \mathbf{A} \mathbf{H}^t)$$

$\mathbf{A}$ adjacency matrix

σ e.g. ReLU



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